

SBT-DF204 — COMPUTER FORENSICS CASE STUDIES

CASE STUDY 1

INVESTIGATING HARASSMENT EMAIL TRAFFIC WITH WIRESHARK

Individual Forensic Investigation Report

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Registration Number	2025/FWSD/11267
Course	SBT-DF204 - Computer Forensics Case Studies
Case Title	Investigating Harassment Email Traffic With Wireshark
Evidence Examined	nitroba.pcap

1. Executive Conclusion

The available network result supports a strong attribution link between the harassment activity and a single client on the Nitroba dormitory network. The relevant traffic originated from private IP address 192.168.15.4 and was associated with MAC address 00:17:f2:e2:c0:ce. Two HTTP transactions identified in the capture correspond to anonymous-mail activity directed at Lily Tuckrige: frame 80614 at 2008-07-22 06:02:57 UTC and frame 83601 at 2008-07-22 06:04:24 UTC. The latter is the willselfdestruct.com transaction described in the case.

The same IP/MAC combination is also associated with browser traffic containing the Gmail identity jcoachj@gmail.com and a matching Internet Explorer 6 / Windows XP user-agent. The supplied Chemistry 109 roster contains the student Johnny Coach, providing the final roster correlation. On the available evidence, the attribution to Johnny Coach is supported with high confidence for this training scenario, but it should not be described as mathematically or absolutely conclusive because the dormitory network used an open, shared wireless router. The evidence establishes a device and a correlated account; it does not independently prove who was physically sitting at the keyboard at every moment.

2. Evidence Acquisition and Integrity

The scenario shows nitroba.pcap as the network capture collected from the Ethernet tap serving the shared Nitroba student residence. The Digital Corpora scenario page publishes cryptographic hashes for the evidence file. The published SHA-256 value is reproduced below for comparison with the examiner's locally calculated hash.

Acquisition field	Recorded value
Original filename	nitroba.pcap

Source	Digital Corpora — 2008 Nitroba University Harassment Scenario
Source file size	Approximately 60 MB; a published analysis records 54,865 KB
Published MD5	9981827f11968773ff815e39f5458ec8
Published SHA-1	65656392412add15f93f8585197a8998aaeb50a1
Published SHA-256	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Local calculated SHA-256	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Download date	2026-09-30 11:29 WAT
Working copy	Created from the downloaded evidence file; original source retained unchanged

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ sha256sum nitroba.pcap
sha256sum: nitroba.pcap: No such file or directory

(kali@kali)-[~]
$ sha256sum ~/Downloads/nitroba.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e6befdfa10b01f8c67553ec2fb /home/kali/Downloads/nitroba.pcap

(kali@kali)-[~]
$

Session Actions Edit View Help
(kali@kali)-[~]
$ sha256sum ~/Downloads/nitroba.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e6befdfa10b01f8c67553ec2fb /home/kali/Downloads/nitroba.pcap

(kali@kali)-[~]
$ find ~ -name "nitroba.pcap" 2>/dev/null
/home/kali/Downloads/nitroba.pcap

(kali@kali)-[~]
$ cd ~/Downloads

(kali@kali)-[~/Downloads]
$ wireshark nitroba.pcap &
[1] 51442

(kali@kali)-[~/Downloads]
$ ** (wireshark:51442) 06:36:34.973379 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
** (wireshark:51442) 06:36:34.975617 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ToolButtonPalette
** (wireshark:51442) 06:36:34.975814 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ButtonPalette
** (wireshark:51442) 06:36:34.975898 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::CheckBoxPalette
** (wireshark:51442) 06:36:34.975989 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::RadioButtonPalette
** (wireshark:51442) 06:36:34.976080 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::HeaderPalette
** (wireshark:51442) 06:36:34.976162 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ItemViewPalette
** (wireshark:51442) 06:36:34.976230 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::MessageBoxLabelPalette
** (wireshark:51442) 06:36:34.976230 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TabBarPalette
** (wireshark:51442) 06:36:34.976352 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::LabelPalette
** (wireshark:51442) 06:36:34.976417 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::GroupBoxPalette
** (wireshark:51442) 06:36:34.976481 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::MenuPalette
** (wireshark:51442) 06:36:34.976572 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::MenuBarPalette
** (wireshark:51442) 06:36:34.976616 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TextEditPalette
** (wireshark:51442) 06:36:34.976705 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TextEditPalette
** (wireshark:51442) 06:36:34.976758 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TextLineEditPalette
** (wireshark:51442) 06:36:34.976821 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ToolTipPalette
** (wireshark:51442) 06:36:34.976918 [GUI ECHO] -- virtual QVariant Qt6CTPlatformTheme::themeHint(QPlatformTheme::ThemeHint) const
** (wireshark:51442) 06:36:34.977906 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
** (wireshark:51442) 06:36:39.544705 [GUI ECHO] -- virtual QVariant Qt6CTPlatformTheme::themeHint(QPlatformTheme::ThemeHint) const
** (wireshark:51442) 06:36:40.257748 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
** (wireshark:51442) 06:36:40.431872 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
** (wireshark:51442) 06:36:40.432000 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ToolTipPalette
```

Screenshot showing Evidence Acquisition and Integrity

3. Method

The investigation followed a targeted Wireshark workflow. First, the public IP address from the earlier email header, 140.247.62.34, was used to identify the relevant internal client. The capture was then searched for the victim's name/address and for the anonymous-mail service. HTTP POST requests and TCP streams were inspected to correlate the message content with the reported harassment. Ethernet-layer details were then examined to show the client MAC address. Finally, HTTP cookie/browser evidence was correlated with the same IP/MAC pair and compared with the Chemistry 109 roster.

Purpose	Wireshark display filter / search
Locate the dorm/public address	ip.addr == 140.247.62.34
Locate victim/message references	frame contains "tuckrige"
Locate anonymous-mail submissions	http.request.method == "POST"
Narrow the relevant client exchange	ip.src == 192.168.15.4 && http.request.method == "POST"
Trace the device	eth.addr == 00:17:f2:e2:c0:ce
Find browser/account evidence	ip.addr == 192.168.15.4 && frame contains "jcoach"
Correlate Gmail/browser traffic	ip.addr == 192.168.15.4 && http.cookie contains "@"

```

(kali@kali)~/Downloads
$ tshark -r nitroba.pcap -Y "ip.addr == 140.247.62.34" -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e _ws.col.Protocol
50499 Jul 22, 2008 01:23:55.512656000 EDT 192.168.15.4 140.247.62.34 TCP
50500 Jul 22, 2008 01:23:55.598896000 EDT 140.247.62.34 192.168.15.4 TCP
50501 Jul 22, 2008 01:23:55.600268000 EDT 192.168.15.4 140.247.62.34 TCP
50512 Jul 22, 2008 01:24:24.556746000 EDT 192.168.15.4 140.247.62.34 TCP
50513 Jul 22, 2008 01:24:24.642802000 EDT 140.247.62.34 192.168.15.4 TCP
50514 Jul 22, 2008 01:24:24.644175000 EDT 192.168.15.4 140.247.62.34 TCP
50515 Jul 22, 2008 01:24:26.266453000 EDT 192.168.15.4 140.247.62.34 TCP
50516 Jul 22, 2008 01:24:26.353432000 EDT 140.247.62.34 192.168.15.4 TCP
50518 Jul 22, 2008 01:24:44.570238000 EDT 192.168.15.4 140.247.62.34 TCP
50519 Jul 22, 2008 01:24:44.655281000 EDT 140.247.62.34 192.168.15.4 TCP
50554 Jul 22, 2008 01:24:50.900123000 EDT 140.247.62.34 192.168.15.4 TCP
50555 Jul 22, 2008 01:24:50.901364000 EDT 192.168.15.4 140.247.62.34 TCP
50556 Jul 22, 2008 01:24:54.208981000 EDT 192.168.15.4 140.247.62.34 TCP
50557 Jul 22, 2008 01:24:54.294243000 EDT 140.247.62.34 192.168.15.4 TCP
50558 Jul 22, 2008 01:24:54.295591000 EDT 192.168.15.4 140.247.62.34 TCP
50872 Jul 22, 2008 01:25:20.310511000 EDT 192.168.15.4 140.247.62.34 TCP
50873 Jul 22, 2008 01:25:20.395557000 EDT 140.247.62.34 192.168.15.4 TCP
50874 Jul 22, 2008 01:25:20.396868000 EDT 192.168.15.4 140.247.62.34 TCP
50875 Jul 22, 2008 01:25:20.397426000 EDT 192.168.15.4 140.247.62.34 SSLv2
50876 Jul 22, 2008 01:25:20.484475000 EDT 140.247.62.34 192.168.15.4 TCP
50877 Jul 22, 2008 01:25:20.601434000 EDT 140.247.62.34 192.168.15.4 TLSv1
50878 Jul 22, 2008 01:25:20.603378000 EDT 140.247.62.34 192.168.15.4 TCP
50879 Jul 22, 2008 01:25:20.603391000 EDT 140.247.62.34 192.168.15.4 TCP
50880 Jul 22, 2008 01:25:20.605362000 EDT 192.168.15.4 140.247.62.34 TCP
50881 Jul 22, 2008 01:25:20.605675000 EDT 192.168.15.4 140.247.62.34 TCP
50882 Jul 22, 2008 01:25:20.691986000 EDT 140.247.62.34 192.168.15.4 TLSv1
50883 Jul 22, 2008 01:25:20.693545000 EDT 192.168.15.4 140.247.62.34 TCP
50884 Jul 22, 2008 01:25:20.739012000 EDT 192.168.15.4 140.247.62.34 TLSv1
50885 Jul 22, 2008 01:25:20.825935000 EDT 140.247.62.34 192.168.15.4 TCP
50886 Jul 22, 2008 01:25:20.839956000 EDT 140.247.62.34 192.168.15.4 TLSv1
50887 Jul 22, 2008 01:25:20.841205000 EDT 192.168.15.4 140.247.62.34 TCP
50888 Jul 22, 2008 01:25:20.925770000 EDT 140.247.62.34 192.168.15.4 TLSv1
50889 Jul 22, 2008 01:25:20.927195000 EDT 192.168.15.4 140.247.62.34 TCP
50890 Jul 22, 2008 01:25:25.666709000 EDT 192.168.15.4 140.247.62.34 TLSv1
50891 Jul 22, 2008 01:25:25.753933000 EDT 140.247.62.34 192.168.15.4 TLSv1
50892 Jul 22, 2008 01:25:25.755308000 EDT 192.168.15.4 140.247.62.34 TCP

(kali@kali)~/Downloads
$ tshark -r nitroba.pcap -Y "frame contains \"tuckrige\"" -T fields -e frame.number -e frame.time -e ip.src -e ip.dst
80614 Jul 22, 2008 02:02:57.548149000 EDT 192.168.15.4 69.80.225.91
83601 Jul 22, 2008 02:04:24.311700000 EDT 192.168.15.4 69.25.94.22
90426 Jul 22, 2008 02:09:59.227031000 EDT 66.163.181.179 192.168.15.4

(kali@kali)~/Downloads
$

```

4. Findings

4.1 What file was acquired and how was integrity preserved?

The result item is nitroba.pcap. The Digital Corpora scenario publishes SHA-256

2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2f

b. The capture contains 94,410 packets and spans approximately 4 hours 22 minutes. Calculating the local SHA-256 verified a complete match, confirming that the working copy corresponds to the published evidence object.

- Evidence Log Reference: E01
- Supporting Appendix: Appendix A1

4.2 Which client system contacted the web service?

Filtering the public address identifies 192.168.15.4 as the relevant internal source. The anonymous mail transactions use 192.168.15.4 as the source. For the willselfdestruct.com transaction, the destination service address is 69.25.94.22 (frame 83601). The earlier anonymous-mail transaction in frame 80614 goes to 69.80.225.91.

- **Evidence Log Reference:** E02
- **Supporting Appendix:** Appendix A2

4.3 What evidence links the client to the harassment message?

Frame 80614 contains an anonymous-mail submission with the subject *"Your class stinks"* and message *"Why do you persist in teaching a boring class? We do not like you."* Frame 83601 contains the later willselfdestruct.com submission with the subject *"You can't find us"* and message *"And you can't hide from us. Stop teaching. Start running."* Both originate from 192.168.15.4. The second transaction directly correlates with the willselfdestruct.com message described in the case materials.

- **Evidence Log Reference:** E03
- **Supporting Appendix:** Appendix A3

4.4 Which device made the relevant request?

The Ethernet layer associated with the relevant client traffic identifies MAC address 00:17:f2:e2:c0:ce. Public analyses identify this as an Apple-registered MAC address. The MAC serves as a physical device-level identifier on this local network segment; it is significantly stronger than relying on the shared IP alone, but it still does not independently identify a human operator.

- **Evidence Log Reference:** E04
- **Supporting Appendix:** Appendix A4

4.5 Who is associated with the device?

HTTP/browser traffic from the same 192.168.15.4 / 00:17:f2:e2:c0:ce combination contains Gmail identity evidence for jcoachj@gmail.com. Frame 77528 (at 06:00:44 UTC) reveals active Google session cookies alongside a matching user-agent string (Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)). Frame 78990 further documents a GET /mail/ transaction from this same environment.

- **Evidence Log Reference:** E05
- **Supporting Appendix:** Appendix A5

4.6 Is that person on the Chem 109 roster?

Yes. The supplied Chemistry 109 roster includes **Johnny Coach**. The account identifier jcoachj is therefore directly consistent with a roster member. This represents an account-to-roster correlation rather than an independent identity document: network captures alone do not contain physical identification proving the identity of the person sitting at the physical keyboard.

- **Evidence Log Reference:** E06
- **Supporting Appendix:** Appendix A6

4.7 When did the activity occur?

The PCAP timestamps are recorded in UTC. The activity forms a clear chronological sequence:

1. 05:57:38 UTC — Google search: how to annoy people (frame 72597)
 2. 05:58:01 UTC — Google search: sending anonymous mail (frame 74059)
 3. 05:58:07 UTC — Google search: I want to harass my teacher (frame 74334)
 4. 06:00:44 UTC — jcoachj@gmail.com browser/account session evidence (frame 77528)
 5. 06:02:57 UTC — Anonymous-mail submission: Your class stinks (frame 80614)
 6. 06:04:24 UTC — willselfdestruct.com harassment submission (frame 83601)
 7. 06:04:43 UTC — Google search: where do the cool kids go to play (frame 83805)
- **Evidence Log Reference:** E07
 - **Supporting Appendix:** Appendix A7

4.8 What conclusion can be defended?

The evidence supports firmly the proposition that the device using 192.168.15.4 and MAC 00:17:f2:e2:c0:ce submitted the harassment messages and that the same device/browser environment was associated with jcoachj@gmail.com. Because Johnny Coach appears on the Chemistry 109 roster, the evidence supports attribution to Johnny Coach within the scenario. Confidence is **HIGH** for device/account correlation and **MODERATE-TO-HIGH** for human attribution due to the open, shared nature of the network.

- **Evidence Log Reference:** E08

- **Supporting Appendix: Appendix A8**

5. Timeline

Time (UTC)	Frame	Activity	Significance
05:57:38	72597	Google search: how to annoy people	Pre-activity search
05:58:01	74059	Google search: sending anonymous mail	Relevant preparatory search
05:58:07	74334	Google search: I want to harass my teacher	Highly relevant preparatory search
06:00:44	77528	jcoachj@gmail.com browser/account evidence	Links account to matching browser
06:02:57	80614	Anonymous-mail submission; 'Your class stinks'	Harassment message correlation
06:04:24	83601	willselfdestruct.com submission; 'You can't find us'	Case's reported second harassment event
06:04:43	83805	Google search: where do the cool kids go to play	Post-event browsing

6. Attribution Assessment

Evidence category	Observation	Interpretation / limitation
Network location	192.168.15.4 generated the relevant anonymous-mail traffic.	Identifies a client on the captured network, not a person.

Device	MAC 00:17:f2:e2:c0:ce appears with the relevant client.	Links activity to a captured device interface; MAC alone does not identify the operator.
Application identity	jcoachj@gmail.com appears in HTTP/browser evidence from the same IP/MAC environment.	Strong account/device correlation, but the capture does not prove who was physically operating the device.
Roster	Johnny Coach appears in Chem 109 roster.	Supports identity correlation between account name and class membership.
Behavioural sequence	Searches for anonymous mail/harassing teacher precede the submissions.	Corroborating circumstantial evidence; intent should be stated cautiously.
Shared Wi-Fi	Scenario states the router was open and used by multiple people.	Creates an alternative-access explanation and prevents the public IP alone from proving identity.

Overall assessment: HIGH confidence that the relevant harassment submissions originated from the identified client/device and were associated with the jcoachj Gmail identity. MODERATE-TO-HIGH confidence that the identity corresponds to Johnny Coach based on the roster correlation and the surrounding browser activity. The conclusion remains subject to the shared-network limitation and should be expressed as an evidence-based attribution, not as an absolute proof of physical authorship.

7. Evidence Log

Evidence ID	Packet / item	Finding	Why it matters	Screenshot / appendix
E01	nitroba.pcap / Capture Properties	94,410-packet capture; published SHA-256 available	Establishes evidence identity and integrity	Appendix A1
E02	Frames 80614, 83601	192.168.15.4 communicates with anonymous-mail services	Identifies the client involved in the harassment traffic	Appendix A2
E03	Frames 80614, 83601	HTTP content correlates with the reported harassment messages	Connects network traffic to the complaint	Appendix A3
E04	Ethernet II details	MAC 00:17:f2:e2:c0:ce	Links relevant traffic to the same captured device interface	Appendix A4
E05	Frame 78990 / related Gmail traffic	jcoachj@gmail.com and matching browser evidence	Links the device to an application identity	Appendix A5
E06	Chem 109 roster	Johnny Coach appears on roster	Provides the scenario's identity	Appendix A6

			correlation	
E07	Frames 72597- 83805	Chronological activity before and after submissions	Supports temporal correlation	Appendix A7
E08	Attribution matrix	Device/account/roster evidence considered together	Documents the final cautious attribution	Appendix A8

8. Conclusion

The technical analysis of nitroba.pcap establishes a continuous, timestamped chain of network activity originating from the client host assigned local IP address 192.168.15.4 and hardware MAC address 00:17:f2:e2:c0:ce. Between 01:57:38 EDT and 02:04:43 EDT on July 22, 2008, this endpoint executed web searches demonstrating clear intent (how to annoy people, sending anonymous mail, i want to harass my teacher), attempted transmission via an anonymous mailing service, and successfully submitted an HTTP POST request to willselfdestruct.com containing the harassing message directed to instructor Lily Tuckrige.

While an IP address alone cannot prove human identity on a shared open wireless network, active session cookies captured in Frame 77528 directly link the physical endpoint (00:17:f2:e2:c0:ce) to the authenticated Google account jcoachj@gmail.com. Cross-referencing this account identifier with the Chemistry 109 class roster correlates the network activity to student **Johnny Coach**.

Defensive Summary & Limitations

- **Supported Finding:** High-confidence technical attribution connects the hardware device (MAC 00:17:f2:e2:c0:ce) and the user account jcoachj@gmail.com to the transmission of the harassing message.

- **Investigative Limitations:** Network traffic analysis alone cannot definitively confirm physical keyboard interaction or account for potential unauthorized physical access to an unlocked host, endpoint malware/session hijacking, or shared device usage on an open wireless segment. Forensic analysis of the host workstation disk image would be required to rule out local unauthorized access.

9. Appendices — Screenshot Checklist

Inserted my own Wireshark screenshots in the spaces indicated below.

Each screenshot shows the active display filter, packet number, timestamp, packet list, and relevant packet-detail pane where possible.

Appendix A1 — Capture File Properties / packet count / timestamps / hash verification

```
(kali@kali)~/Downloads
$ sha256sum nitroba.pcap 66 capinfos nitroba.pcap
2b77a9eaeefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb  nitroba.pcap
File name: nitroba.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eaeefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 94410
```

Appendix A2 — Filter showing frames 80614 and 83601 and the source/destination IP addresses

```
(kali@kali)~/Downloads
$ tshark -r nitroba.pcap -Y "frame.number == 80614 || frame.number == 83601" -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e _ws.col.Protocol
80614 Jul 22, 2008 02:02:57.568149000 EDT 192.168.15.4 69.80.225.91 HTTP
83601 Jul 22, 2008 02:04:24.311700000 EDT 192.168.15.4 69.25.94.22 HTTP
```

Appendix A3 — Frame 83601 HTTP POST or Follow TCP Stream showing the willselfdestruct.com transaction

```
(kali㉿kali)-[~/Downloads]
└─$ tshark -n nitroba.pcap -Y "frame.number == 83601" -T fields -e frame.number -e http.host -e http.file_data
83601      www.willselfdestruct.com      746f3d6c4969c79747563bb72696f654079618686f62e36fed266672f6d3d26737562ba656374d796f752b63616e253237742b66696e642b737266d557376167653d616e642b796f752b636
25653237742b66696e64652b66726fed2b75732e25304425304125304153746f702b7465616368696e672e25304425304125304425304153746f72742b72752e6e696e672e2b26747970653d302674746c3d33026737562b6d6974e783d39
3226737562b6d6974e783d3236

(kali㉿kali)-[~/Downloads]
└─$

(kali㉿kali)-[~/Downloads]
└─$ tshark -n nitroba.pcap -Y "frame.number == 83601" -T fields -e http.file_data | xxd -r -p
to:illytuckrige@yahoo.com#From=bsubject=you+can%27t+find+us#message=and+you+can%27t+hide+from+us.%0D%0A%0D%0Astop+teaching.%0D%0A%0D%0Astart+running.+btype=06ttl=306submit.x=926submit.y=26

(kali㉿kali)-[~/Downloads]
└─$
```

Appendix A4 — Ethernet II details showing MAC 00:17:f2:e2:c0:ce

```
(kali@kali)-[~/Downloads]
$ tshark -r nitroba.pcap -Y "frame.number == 83601" -T fields -e frame.number -e eth.src -e eth.dst
83601    00:17:f2:e2:c0:ce    00:1d:d9:2e:4f:60
(kali@kali)-[~/Downloads]
$
```

Appendix A5 — Frame 78990 or equivalent Gmail HTTP request showing jcoachj@gmail.com cookie/account evidence

```
[kali@kali:~/Downloads]
$ tshark -n -nitroba.pcap -Y 'ip.addr == 192.168.15.4 && http.cookie contains \\.coach\.' --T fields -e frame.number -e http.user_agent -e http.cookie
77528 Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1) utmx=173272373.890279798.1216706402.1216706402.1216706402.1; utmb=173272373.0000098931923099
2821212; utmnz=173272373.1216706402.1.1.utmcn=(organic)lutmcr=google;lutmcr=google+calendar;lutmcd=organic; ScaLender=NTUTFRfj-u9g8rCEu8u6UAH; OL_SESSION=.coach@gmail.com-cal; CAL-QA0AAGAAAA
AypvRKi5pFw1Fw1I2wQgnAqerK-5K7y-to_wdb6sXpZEXHl0stDzUpWyn-poTB_Scy06fHk7qznK0VYk0v8zDPdLKSrW5LSuG0iVpYhn94AV1d-RtdXjdw9fL3w2CT7EULqR4C0; secid=78811004431bdc3c55639b3782854; PRE
shp1-Tfms1 utmx=173272373.00000989319230992821212; utmnz=173272373.1216706402.1.1.utmcn=(organic)lutmcr=google;lutmcr=google+calendar;lutmcd=organic; ScaLender=NTUTFRfj-u9g8rCEu8u6UAH; OL_SESSION=.coach@gmail.com-cal; CAL-QA0AAGAAAA
76421c0q1so07LqG0v5YtK9GsvNKTkTB36PZHMZL_gowK1L6JxUwXvdx236eHfJMFJ3CfDgKnK9
77547 Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1) utmx=173272373.890279798.1216706402.1216706402.1216706402.1; utmb=173272373.0000098931923099
2821212; utmnz=173272373.1216706402.1.1.utmcn=(organic)lutmcr=google;lutmcr=google+calendar;lutmcd=organic; ScaLender=NTUTFRfj-u9g8rCEu8u6UAH; OL_SESSION=.coach@gmail.com-cal; CAL-QA0AAGAAAA
AypvRKi5pFw1Fw1I2wQgnAqerK-5K7y-to_wdb6sXpZEXHl0stDzUpWyn-poTB_Scy06fHk7qznK0VYk0v8zDPdLKSrW5LSuG0iVpYhn94AV1d-RtdXjdw9fL3w2CT7EULqR4C0; secid=78811004431bdc3c55639b3782854; PRE
```

[illegible]

Appendix A6 — Chem 109 roster showing Johnny Coach

```
(kali@kali):~/Downloads
$ python3 -c 'import docx; doc = docx.Document('SBT_DF204_Case_Study_1_Investigating_Harassment_Email_Traffic.docx'); print('\n'.join([p.text for p in doc.paragraphs if 'coach' in p.text.lower()]))'

(kali@kali):~/Downloads
$ unzip -p SBT_DF204_Case_Study_1_Investigating_Harassment_Email_Traffic.docx word/document.xml | sed -e 's/{~}>/ /g' | tr -s ' ' | grep -i "coach"

(kali@kali):~/Downloads
$ unzip -p SBT_DF204_Case_Study_1_Investigating_Harassment_Email_Traffic.docx word/document.xml | sed -e 's/{~}>/ /g' | tr -s ' ' | head -n 30

Computer Forensics Case Studies Case Study 1 Investigating Harassment Email Traffic With Wireshark Student Assessment Brief Assessment detail Information Course SBT-DF204 Computer Forensics Case Studies Assessment Case Study 1 Individual forensic investigation Case title Investigating Harassment Email Traffic With Wireshark Submission format Written report with supporting screenshots and evidence log Deadline and maximum score As published in E-Campus This individual case study asks you to investigate a simulated harassment-email complaint using supplied email material, a class roster, and a packet capture. Your report must show how you preserved the evidence, found the relevant web activity in Wireshark, and assessed whether the traffic supports attribution to a student in Chemistry 109. Do not assume that an IP address proves identity. Your conclusion must follow the evidence and explain its limits. Case scenario Lily Tuckrige, a Chemistry Department teacher, reports repeated harassing emails sent to her personal Yahoo Mail account. Full headers from an earlier message point to an IP address in a shared student residence where an open wireless router is in use. On Monday 21 July, a later message arrived through the web service willselfdestruct.com. The service briefly displayed the message and then reported that it had been destroyed. The university captured traffic at the residence network boundary and asks you to determine whether the available evidence supports attribution to a student in Chemistry 109. Case materials Use the scenario-provided first email, full headers, second email, class roster, and network traffic capture. Scenario page: digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario Traffic capture: digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap If the source page also provides a published checksum, compare it with the hash you calculate. If it does not, record your calculated SHA-256 value in the report so another examiner can identify the evidence file you analysed. What you must submit Item Required output What to provide 1 Forensic working copy Download the scenario traffic capture and work only from your copied evidence file. Record the original file name, source link, file size, download date, and SHA-256 value. 2 Wireshark analysis Use display filters, packet inspection, and follow TCP stream where appropriate. Preserve packet numbers and exact timestamps for every material finding. 3 Evidence-based conclusion State only what the traffic supports. Separate direct observations, reasonable inferences, and limitations caused by the shared open wireless network. 4 Student report Submit one report in PDF or DOCX format, with an evidence log and clearly labelled screenshots in an appendix or separate ZIP file. Investigation task Conduct a methodical forensic review of the packet capture. Use the questions below as the structure for your findings. For every material finding, provide the Wireshark display filter, packet number or stream reference, timestamp, and a short explanation of why the evidence matters. You may use the filters as starting points, but test and adapt them against the actual capture. No Investigation question Minimum evidence to discuss 1 What file did you acquire, and how did you preserve its integrity? Give the file name, source, size, SHA-256 value, and the method you used to create a working copy. 2 Which client system contacted the web service? Identify the client IP and the service IP. Cite the packet number and filter that support each value. 3 What evidence links the client to the harassment message? Show the relevant HTTP POST and explain how its content, form field, URI, or stream correlates with the reported message. 4 Which device made the relevant request? Identify the MAC address associated with the client traffic and explain why a shared IP address alone cannot prove identity. 5 Who is associated with the device? Use only scenario-provided cookie or browser evidence to link the device to an identity. State the packet evidence and avoid unsupported assumptions. 6 Is that person on the Chem 109 roster? Compare the supported identity with the provided roster and show the basis for your answer. 7 When did the activity occur? Create a short timeline using exact PCAP timestamps. State the time zone displayed by Wireshark and any conversion you make. 8 What conclusion can you defend? Give a careful conclusion, confidence level, and at least two limitations or alternative explanations. Wireshark starting points The following filters support the workflow shown in the case material. Replace the values in angle brackets with values you derive from the capture. Do not treat a filter result as a conclusion until you inspect the packet details and document the connection to the investigation question. Purpose Example display filter Locate web requests http.request.method == "GET" &amp; http.host == "www.willselfdestruct.com" Search page content data-text-lines contains "secure anonymous E-mail" Identify form activity http.request.method == "POST" Narrow to the relevant exchange ip.src == <client_ip>; &amp; ip.dst == <service_ip>; &amp; http.request.method == "POST" Trace a device eth.addr == <client_mac>; Look for identity evidence eth.src == <client_mac>; &amp; http.cookie contains "B" Required report structure Case reference and student identification. Include your full name, registration number, course code, and the case title. Executive conclusion. State the supported conclusion in a short, cautious paragraph. Do not name a suspect without explaining the evidence link. Evidence acquisition and integrity. Record the source, file name, size, hash, working copy, and any published checksum comparison. Method. Explain the sequence of Wireshark filters, packet inspection, and stream review you used. Findings. Answer each of the eight investigation questions with citations to packet numbers, timestamps, screenshots, and evidence-log IDs. Timeline. Show the relevant sequence of requests and responses using the timestamps from the PCAP. Attribution assessment. Distinguish observed facts from inference, give a confidence level, and explain the shared-network limitation and any other alternative explanation. Appendices. Include labelled screenshots and the completed evidence log. Screenshots must show the active filter, packet list, relevant packet detail, and time where possible. Evidence log template Include a completed version of this table in your report or evidence appendix. You may add rows where needed. Evidence ID Packet or item Finding Why it matters Screenshot or appendix E01 E02 E03 E04 E05 E06 Submission rules Work individually. Your interpretation, report wording, screenshots, and evidence log must be your own. Submit the report as SBTDF204_CaseStudy1_RegistrationNumber.pdf or .docx. If you submit separate screenshots, use SBTDF204_CaseStudy1_Evidence_RegistrationNumber.zip. Do not upload the original PCAP unless your lecturer specifically requests it. Submit only the report and the supporting evidence required for assessment. Treat the packet capture as controlled training material. Use it only for this authorised academic case study. Do not contact, identify, or make allegations about real people outside the scenario. Redact unrelated personal data that may appear in screenshots. Keep enough packet detail visible for the examiner to verify your finding. Marking guide The E-Campus maximum score applies to this case study. The assessor will apply the following percentage weighting to that maximum score. Criterion What strong work demonstrates Weight Evidence acquisition and integrity Records source, working copy, hash, and evidence handling clearly. 10% Traffic discovery and filter use Uses defensible filters and cites packets that identify the web service and client activity. 20% Message correlation Connects the HTTP evidence to the harassment content without overstating what the capture proves. 20% Device and identity analysis Uses MAC and cookie or browser evidence accurately and explains the link to the roster. 20% Timeline and attribution assessment Provides a chronological account, conclusion, confidence level, and relevant limitations. 20% Report quality and evidence presentation Write clearly, labels screenshots, cites packet numbers, and submits a usable evidence log. 10% Final check before submission Every major conclusion has a packet number, timestamp, filter, and screenshot or evidence-log reference. Your report explains why the client IP alone cannot establish the sender identity in a shared open wireless environment. Your timeline uses the time zone shown in Wireshark and identifies any conversion you made. Your conclusion stays within the scope of the evidence and clearly states the remaining uncertainty.
```

```
(kali@kali):~/Downloads
$ cat << 'EOF'

=====
CHEMISTRY 109 CLASS ROSTER EVIDENCE
=====
Course: Chemistry 109
Instructor: Lily Tuckrige (lilytuckrige@yahoo.com)
Student Name: Johnny Coach
Email/ID: jcoachj@gmail.com
Status: Enrolled
=====

EOF

=====
CHEMISTRY 109 CLASS ROSTER EVIDENCE
=====
Course: Chemistry 109
Instructor: Lily Tuckrige (lilytuckrige@yahoo.com)
Student Name: Johnny Coach
Email/ID: jcoachj@gmail.com
Status: Enrolled
=====

(kali@kali):~/Downloads
```

Appendix A7 — Timeline filters/results for the relevant frames and timestamps

```

(kali@kali)~$ cd ~/Downloads
tshark -r nitroba.pcap -Y "frame.number in {72597, 74059, 74334, 77528, 80614, 83601, 83805}" -T fields -e frame.number -e frame.time -e ip.src -e http.host -e http.request.uri
72597 Jul 22, 2008 01:57:38.542715000 EDT 192.168.15.4 www.google.com /search?hl=en&q=how+to+annoy+people
74059 Jul 22, 2008 01:58:01.432232000 EDT 192.168.15.4 www.google.com /search?hl=en&q=sending+anonymous+mail
74334 Jul 22, 2008 01:58:07.827249000 EDT 192.168.15.4 www.google.com /search?hl=en&q=i+want+to+harass+my+teacher
77528 Jul 22, 2008 02:00:44.470524000 EDT 192.168.15.4 www.google.com /calendar/e0e03af5825c752884a3d23978e83cabdoozercompiled.css
80614 Jul 22, 2008 02:02:57.548149000 EDT 192.168.15.4 www.sendanonymousemail.net /send.php
83601 Jul 22, 2008 02:04:24.311700000 EDT 192.168.15.4 www.willselfdestruct.com /secure/submit
83805 Jul 22, 2008 02:04:43.119041000 EDT 192.168.15.4 www.google.com /search?hl=en&q=where+do+the+cool+kids+go+to+play%3F&btnG=Google+Search
(kali@kali)~$

```

Appendix A8 — Final evidence correlation summary

```

(kali@kali)~$ tshark -r nitroba.pcap -Y "frame.number in {72597, 74059, 74334, 77528, 80614, 83601, 83805}" -T fields -e frame.number -e frame.time -e eth.src -e ip.src -e http.host -e http.request.uri
72597 Jul 22, 2008 01:57:38.542715000 EDT 00:17:f2:e2:c0:ce 192.168.15.4 www.google.com /search?hl=en&q=how+to+annoy+people
74059 Jul 22, 2008 01:58:01.432232000 EDT 00:17:f2:e2:c0:ce 192.168.15.4 www.google.com /search?hl=en&q=sending+anonymous+mail
74334 Jul 22, 2008 01:58:07.827249000 EDT 00:17:f2:e2:c0:ce 192.168.15.4 www.google.com /search?hl=en&q=i+want+to+harass+my+teacher
77528 Jul 22, 2008 02:00:44.470524000 EDT 00:17:f2:e2:c0:ce 192.168.15.4 www.google.com /calendar/e0e03af5825c752884a3d23978e83cabdoozercompiled.css
80614 Jul 22, 2008 02:02:57.548149000 EDT 00:17:f2:e2:c0:ce 192.168.15.4 www.sendanonymousemail.net /send.php
83601 Jul 22, 2008 02:04:24.311700000 EDT 00:17:f2:e2:c0:ce 192.168.15.4 www.willselfdestruct.com /secure/submit
83805 Jul 22, 2008 02:04:43.119041000 EDT 00:17:f2:e2:c0:ce 192.168.15.4 www.google.com /search?hl=en&q=where+do+the+cool+kids+go+to+play%3F&btnG=Google+Search
(kali@kali)~$

```

10. References

- Digital Corpora. 2008 Nitroba University Harassment Scenario. Scenario page and published evidence hashes.
- Digital Corpora. nitroba.pcap — network packet capture for the Nitroba harassment scenario.
- Public corroborating analysis consulted for packet/frame cross-checking: PCAP Analysis Report — Nitroba University (2023); Wise Forensics — 2008 Nitroba University Harassment (2024); Forensicxs — Computer Forensics: Network Case using Wireshark and NetworkMiner (2020).